

checking optimality of the solution using
Modi method / UV method

5 ⁽⁶⁾	1 ⁽⁶⁾	8
9 ⁽³⁾	4	0 ⁽¹¹⁾
17	6 ⁽⁴⁾	7

$u_1 = 0$ given in the solution
for the problem
discussed in the
previous chapter.

Consider allocated cells $C_{ij} = u_i + v_j$

$$u_1 = 0 \quad v_1 = 5$$

$$u_2 = 4 \quad v_2 = 1$$

$$u_3 = 5 \quad v_3 = -4$$

$$C_{11} = u_1 + v_1$$

$$5 = 0 + v_1 \Rightarrow v_1 = 5$$

$$C_{12} = u_1 + v_2$$

$$1 = 0 + v_2 \Rightarrow v_2 = 1$$

$$C_{21} = u_2 + v_1$$

$$9 = u_2 + 5 \Rightarrow u_2 = 4$$

$$C_{23} = u_2 + v_3$$

$$0 = 4 + v_3 \Rightarrow v_3 = -4$$

$$C_{32} = u_3 + v_2$$

$$6 = u_3 + 1 \Rightarrow u_3 = 5$$

checking opportunity cost of the unallocated cells

9

$$\Delta_{ij} = C_{ij} - (u_i + v_j)$$

$$\Delta_{13} = C_{13} - (u_1 + v_3) \Rightarrow 8 - (0 + -4) \Rightarrow +12$$

$$\Delta_{22} = C_{22} - (u_2 + v_2) \Rightarrow 4 - (4 + 1) \Rightarrow -1 \longrightarrow \text{(-ve)}$$

$$\Delta_{31} = C_{31} - (u_3 + v_1) \Rightarrow 17 - (5 + 5) \Rightarrow +7$$

$$\Delta_{33} = C_{33} - (u_3 + v_3) \Rightarrow 7 - (5 + -4) \Rightarrow +6$$

there is only one negative $\Delta_{22} = -1$ in the cell C_{22}

Assign +ve to this cell C_{22}

C_{22} & C_{11} are -ve cells

C_{12} & C_{21} are +ve cells

Consider circled values in all cells,

In the -ve cells, least value is 3

↓
Add to +ve cells
subtract in -ve cells

$$6 - 3 = 3$$

$$3 - 3 = 0 \text{ (null)}$$

$$6 + 3 = 9$$

$$0 + 3 = 3$$

5	1	8
9	4	0
17	6	7

The new cell becomes

5	1	8	12
9	4	0	14
17	6	7	4
9	10	11	30

Allocated cells = 5
 $m+n-1 = 3+3-1 = 5$ } one solution is nondegenerate

Initial feasible solution

$$= (9 \times 5) + (3 \times 1) + (3 \times 4) + (11 \times 0) + (4 \times 6) = 84$$

Checking for optimality for the allocated cells
(using modi method)

5 ^⑨	1 ^③	8
9	4 ^③	0 ^⑪
17	6 ^④	7

$u_1 = 0$ Allocated cells $C_{ij} = u_i + v_j$

$$C_{11} = u_1 + v_1$$

$$5 = 0 + v_1 \Rightarrow v_1 = 5$$

$$C_{12} = u_1 + v_2$$

$$1 = 0 + v_2 \Rightarrow v_2 = 1$$

$$C_{22} = u_2 + v_2$$

$$4 = u_2 + 1 \Rightarrow u_2 = 3$$

$$C_{23} = u_2 + v_3$$

$$0 = 3 + v_3 \Rightarrow v_3 = -3$$

$$C_{32} = u_3 + v_2$$

$$6 = u_3 + 1 \Rightarrow u_3 = 5$$

Checking the

unallocated cells

$$\Delta_{ij} = C_{ij} - (u_i + v_j)$$

$$\Delta_{13} = C_{13} - (u_1 + v_3) = 8 - (0 + -3) = +11$$

$$\Delta_{21} = C_{21} - (u_2 + v_1) = 9 - (3 + 5) = +1$$

$$\Delta_{31} = C_{31} - (u_3 + v_1) = 17 - (5 + 5) = +7$$

$$\Delta_{33} = C_{33} - (u_3 + v_3) = 7 - (5 + -3) = +5$$

All values are +ve.

Initial feasible solution 84

is the optimal and unique solution